

PRODUCT ANALYTICS PROJECT

A comprehensive study of a 5,000-session dataset for a mature e-commerce application. This report evaluates five key hypotheses regarding user funnel friction, time-spent variances, device-specific UI discrepancies, traffic acquisition sources, and interface button layout according to UX laws.

TABLE OF CONTENTS

- 01

PROJECT INTRODUCTION & CONTEXT
Overview of the e-commerce dataset, objectives of the drop-off analysis, and details regarding homepage session initiation.
- 02

USER FUNNEL & HYPOTHESIS 1
Analyzing the user funnel conversion rates (Home to Confirmation) and evaluating Hypothesis 1 (low intent vs window shopping) using average time-on-page data.
- 03

DEVICE DISCREPANCIES & FEATURE ENGAGEMENT
Evaluating Hypothesis 2 (UI inconsistencies across Mobile, Tablet, and Desktop violating Jakob's Law) and Hypothesis 3 (feature engagement of window shoppers).
- 04

TRAFFIC REFERRAL SOURCE INTENT
Testing Hypothesis 4 regarding intent variances for users redirected from external sources (Direct, Email, Google, Social Media) using referral source drop-off stats.
- 05

USER INTERFACE FRICTION & UX LAWS
Investigating Hypothesis 5 (application UI friction) by evaluating target button size and placement according to Fitts's Law and Jakob's Law.
- 06

RECOMMENDED NORTH STAR METRIC FRAMEWORK
Establishing the primary North Star Metric (NSM), supporting L1 metrics, and counter-metrics for a mature-stage e-commerce company.

PROJECT CONTEXT

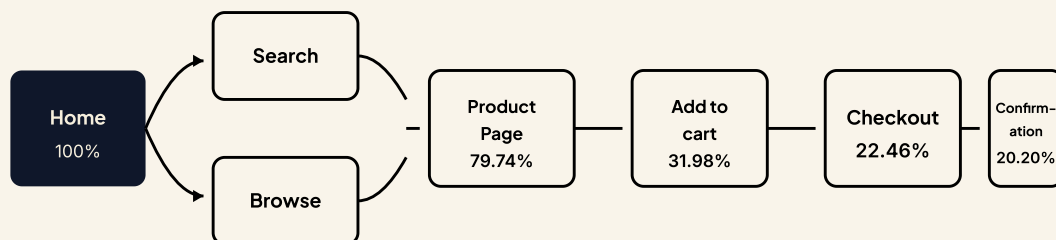
We have a dataset of 5000 unique sessions of people worldwide on a particular e commerce application.

This project is catered to understand where the application is facing the most amount of drop off in users and as to how we could renovate the product to enhance its revenue.

Starting with the funnel we have a total of 5000 unique session which are being initiated from the homepage.

Surprisingly even if the users are not being driven organically but rather through redirects from social media, mail or google their funnel still starts at the homepage - *"Perks of synthetic data I suppose"*.

USER FUNNEL DIAGRAM



As you can see the biggest drop off amongst users occur during product page $\Rightarrow$ add to cart ie about a 47% drop off therefore lets investigate that.

HYPOTHESIS EVALUATION

HYPOTHESIS – 1: LOW CUSTOMER INTENT

Most of the window shoppers or low intent users prefer to check the product pages of several accessories and then close the application rather than adding them to the cart as it is a high commitment action.

This hypothesis can be validated by checking the avg time spent by the non paying user on the product page which should be significantly smaller than the paying users as the low intent customers just check the product rather than getting into its specification unlike the high intent customers and therefore would spend lesser time on the page.

The average(time) spent by users who make and don't make the purchase is about the same therefore proving our hypothesis wrong that the major reason of drop off is low intent customers, Low customer intent might be a minor reason but the major underlying reason seems to be something else.

PURCHASED	AVG(TIMEONPAGE_SECONDS)
0	97.42761168962042
1	95.96534653465346

HYPOTHESIS – 2: UI DISCREPANCY ACROSS DEVICES

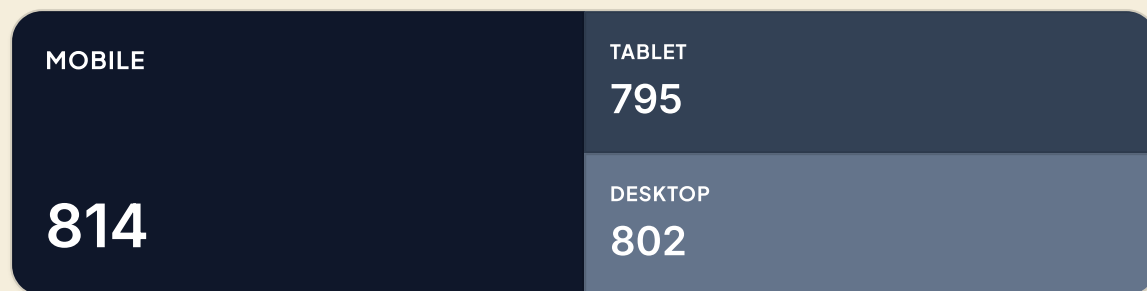
UI discrepancy across devices, Different UI amongst different devices caused friction in user experience as it violates Jakob's law which states that user believes that your application behaves the same way other applications which they use, such as if the ui elements are situated elsewhere such as the add to cart icon on the tablet as compared to the other devices then the users might have trouble proceeding to the next step.

We can simply validate this hypothesis by checking if there is a major variation in the percentage of non paying users when grouped by device, ie some device having an absurdly high number while the other having a very low percentage.

After plotting the graph its clearly visible that the non paying users are distributed equally amongst all three platforms therefore even if its a UI issue it isnt specific to a single device.

DEVICE & ENGAGEMENT METRICS

WINDOW SHOPPERS DISTRIBUTION BY DEVICE



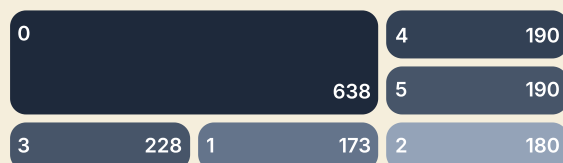
HYPOTHESIS- 3: FEATURE ENGAGEMENT

The third hypothesis builds on to the first hypothesis of most of the users being low intent customers, A window shopper browses several products but has low intention of buying them therefore their engagement with other features should be less.

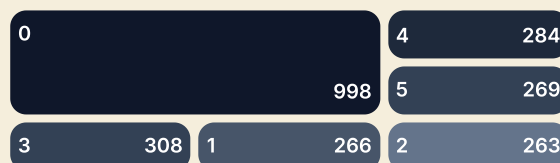
We can validate this by checking the total number of users who don't move on from the product page to the cart page and check if they don't add anything to the cart ie interact with the add to cart button.

We see that the percentage of people not interacting with the add to cart button is almost the same in both paying and non paying users that is about 40% therefore proving this hypothesis wrong as well.

USERS WHO CONTINUE TO CART PAGE



USERS WHO DROP OFF AT PRODUCT PAGE

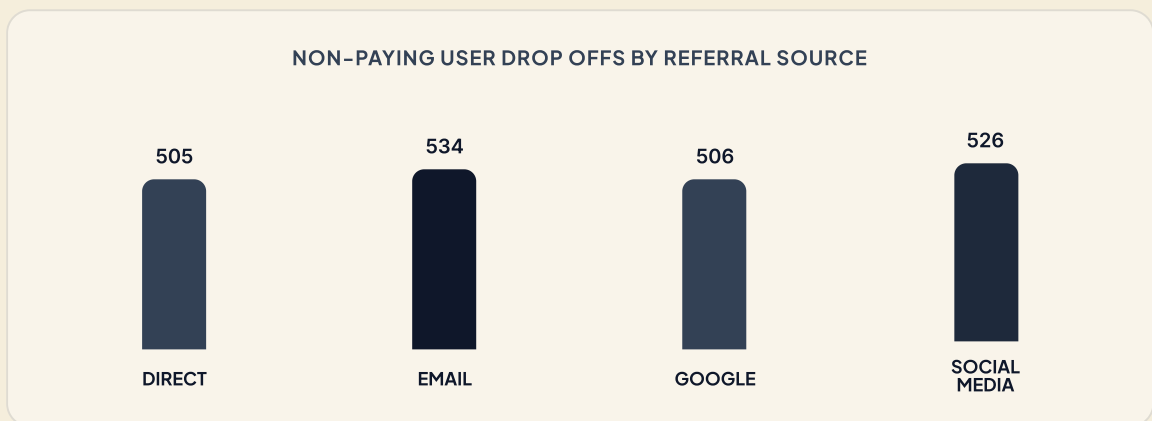


TRAFFIC ACQUISITION ANALYSIS

HYPOTHESIS-4: REFERRAL SOURCE INTENT

It's also possible that the users who have been redirected by external sources such as social media or email have lesser intent of buying the product as they might have been redirected by mistake rather than them actually being interested in the product, which happens to users when they are parsing through social media or emails.

This can simply be validated by checking the percentage of non paying customers who have arrived at the product page through emails and social media. If these numbers are significantly higher than the non paying customers who have either come directly or through google search then our hypothesis is correct.



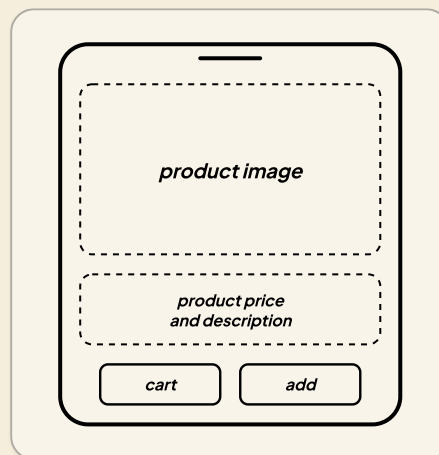
From the graph its clearly visible that the number of non paying consumers are redirected more from emails and social media but the variation is of only about 25 customers on average and therefore making this variation more or less insignificant, therefore this might be one of the minor reason but it isn't the major underlying reason for the drop off, hence this hypothesis is wrong as well.

INTERFACE & LAYOUT ANALYSIS

HYPOTHESIS-5: UI DESIGN AND FLOW FRICTION

One of the reasons that we have ignored till now is that the problem might not be low intent customers but rather the UI of the application itself across all devices, from the data its visible that both the paying and non paying customers spend about the same amount of time on the product pages, display similar engagement with other features and similar data as to from which platform they have been referred from, therefore its possible that there might be friction in the user experience in going from product page $\Rightarrow$ cart that might be causing this drop off.

This hypothesis is harder to validate due to the lack of data about the UI of the application but from the data its visible that about 40% of both paying and non paying consumers don't engage with the add to cart button in the product page therefore it is possible that the cluttered UI might be the reason of this drop off.



According to UX principles such as Jakob's law and Fitt's law the add to cart and the cart button should both be at a place that they are in other e commerce applications and at the same time should be large enough as size of the target/icon is inversely proportional to the time taken to click the icon therefore if the platform is violating either of these laws it might have caused a drop off amongst users and as seeing that our other hypothesis have already been proved wrong and as this hypothesis is slightly validated we consider this to be the root cause of the drop off.

RECOMMENDED PRODUCT METRICS

METRICS TO BE MEASURED

As we have been provided data about the user funnel and not about acquisition its fair to assume that the Product is in its mature stage, therefore our focus lies on engagement and monetisation rather than acquiring new users.

NORTH STAR METRIC (NSM)

NUMBER OF ORDERS PLACED PER DAY PER USER

The primary metric representing success for an E-commerce company in its mature stage, aligning both user value and business revenue.

L1 SUPPORTING METRICS

Supporting metrics should be **ctr (click through rate) of add to cart**, **ctr of cart icon** and other **ctas (Click through actions)** at the product page.

As the biggest drop off is at the product page therefore if the engagement metrics of product page rise it would signify rise in the users going through the funnel therefore rise in orders placed.

COUNTER METRICS

Some counter metrics that we need to consider are **Total number of orders placed** and **total number of users** - as there is a chance that the user base might fall but the NSM will go up which might mislead us.